

TrES-3b

R-0390

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_180607 · created 2026-07-19T00:04:55+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458276.8841 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.111 +/- 0.029
Transit depth (Rp/Rs)^2	1.23 +/- 0.65 [%]
Orbital Inclination (inc)	82.95 +/- 1.66
Airmass coefficient 1 (a1)	1.059 +/- 0.011
Airmass coefficient 2 (a2)	-0.052 +/- 0.026
Scatter in the residuals of the lightcurve fit is	1.08 %
Best Comparison Star	#3 - [339, 28]
Optimal Aperture	3.83
Optimal Annulus	9.84
Transit Duration (day)	0.055 +/- 0.016

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.111 ± 0.029 vs 0.16309 (deviation 1.65σ)
Duration fit vs published	0.055 d vs 0.0567 d (deviation 0.1σ)
Mid-time O–C	-4.3 min
Depth SNR	3.5
Residual scatter	1.08 %

Light curve

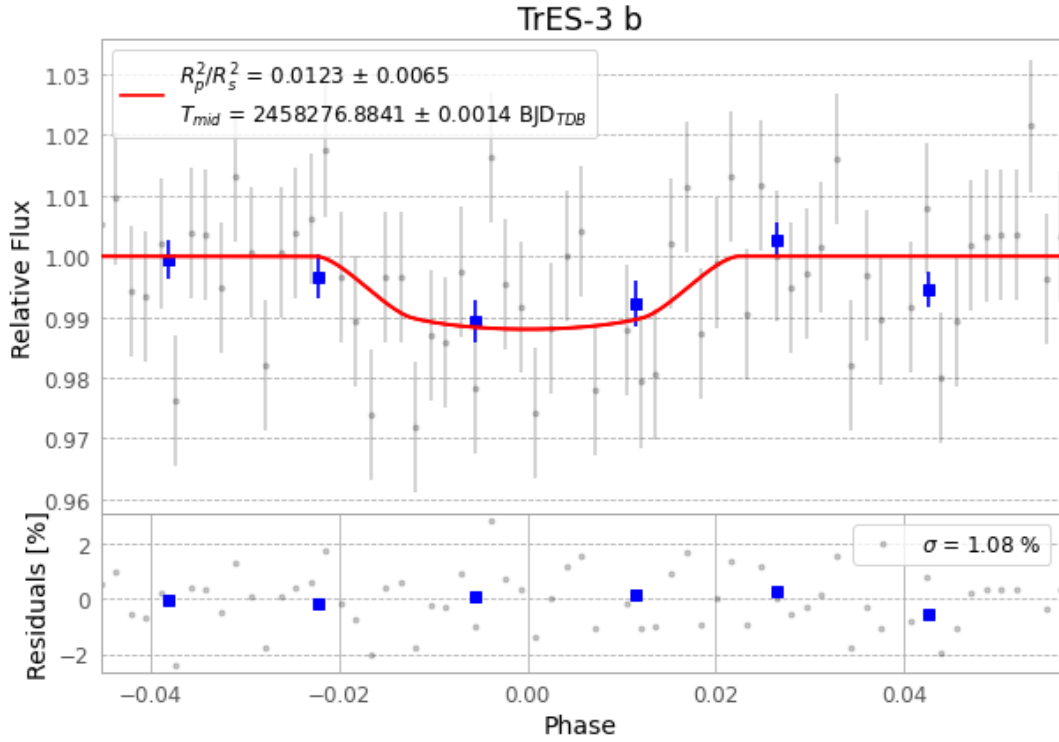


Figure 1 — FinalLightCurve_TrES-3 b_2018-06-07.png (SHA-256 ba0c32648c04e5a7...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [372, 243], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 3f9c4c3098ed8202...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 8350c6b

Data provenance

65 FITS frames (43 MB), TRES-3180607074212.FITS → TRES-3180607105412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-180607.log (2cc0c8ea5244...), FinalLightCurve_TrES-3 b_2018-06-07.png (ba0c32648c04...), FinalLightCurve_TrES-3 b_2018-06-07.csv (132a427776af...)

Compute resources

Wall time 140.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-19 00:04Z.